

PROJECT AND TEAM INFORMATION

Project Title

HIDS: Intelligent Host Security Monitoring System

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Modern computer systems generate a large amount of activity through user logins, file changes, running processes, and network connections. Some of these activities may indicate unauthorized access or an ongoing attack. Manually checking all these activities is difficult and important security events can easily be missed.

A Host-Based Intrusion Detection System (HIDS) monitors activities happening on an individual computer and looks for suspicious or unusual behavior. It can detect events such as repeated failed login attempts, unexpected file modifications, suspicious processes, and unusual network activity.

Our project will develop a simplified HIDS that collects host-level security events, analyzes them using predefined rules, and generates alerts when suspicious activity is detected. The system will also maintain security logs so that detected events can be reviewed later.

We chose this problem because it is a practical cybersecurity application where the DSA and OOP concepts learned in the course can be applied together. Data structures can be used to organize and process security events, while OOP can be used to represent different monitoring and detection components.

State of the Art / Current solution

Currently, computer systems use different security tools such as antivirus software, firewalls, system logs, and security monitoring applications to identify potential threats. Operating systems also maintain logs related to user authentication, processes, and other system activities.

Enterprise security environments may use Security Information and Event Management (SIEM) systems and other monitoring tools to collect and analyze security events. These solutions can provide advanced detection and centralized monitoring, but they can also be complex and are often designed for larger environments.

Our project will provide a simplified HIDS focused on understanding the basic principles of host-level intrusion detection. Instead of developing a complete commercial security product, the system will simulate or monitor selected host activities and apply predefined detection rules.

The project will demonstrate how DSA concepts such as queues, hash maps, sets, searching, sorting, and priority queues can be used to manage and analyze security events.

Project Goals and Milestones

The main goal of our project is to develop a Host-Based Intrusion Detection System that can monitor selected activities on a computer, identify suspicious behavior, and generate security alerts.

Our main goals are to:

Monitor user authentication and login events. Detect repeated failed login attempts. Monitor important files for unexpected modifications or deletion. Track running processes and identify suspicious or unknown processes. Monitor selected network connection information. Store and organize security events efficiently. Assign different severity levels to detected events. Generate alerts when predefined suspicious activity is detected. Maintain a security event history for later analysis. Apply DSA and OOP concepts throughout the system.

We will first identify the types of security events and define the main classes required for the system. We will then implement the individual monitoring and detection components. After that, DSA structures and detection rules will be integrated with the OOP design. Finally, we will test the complete system using different normal and suspicious activity scenarios and evaluate whether the expected alerts are generated.

Project Approach

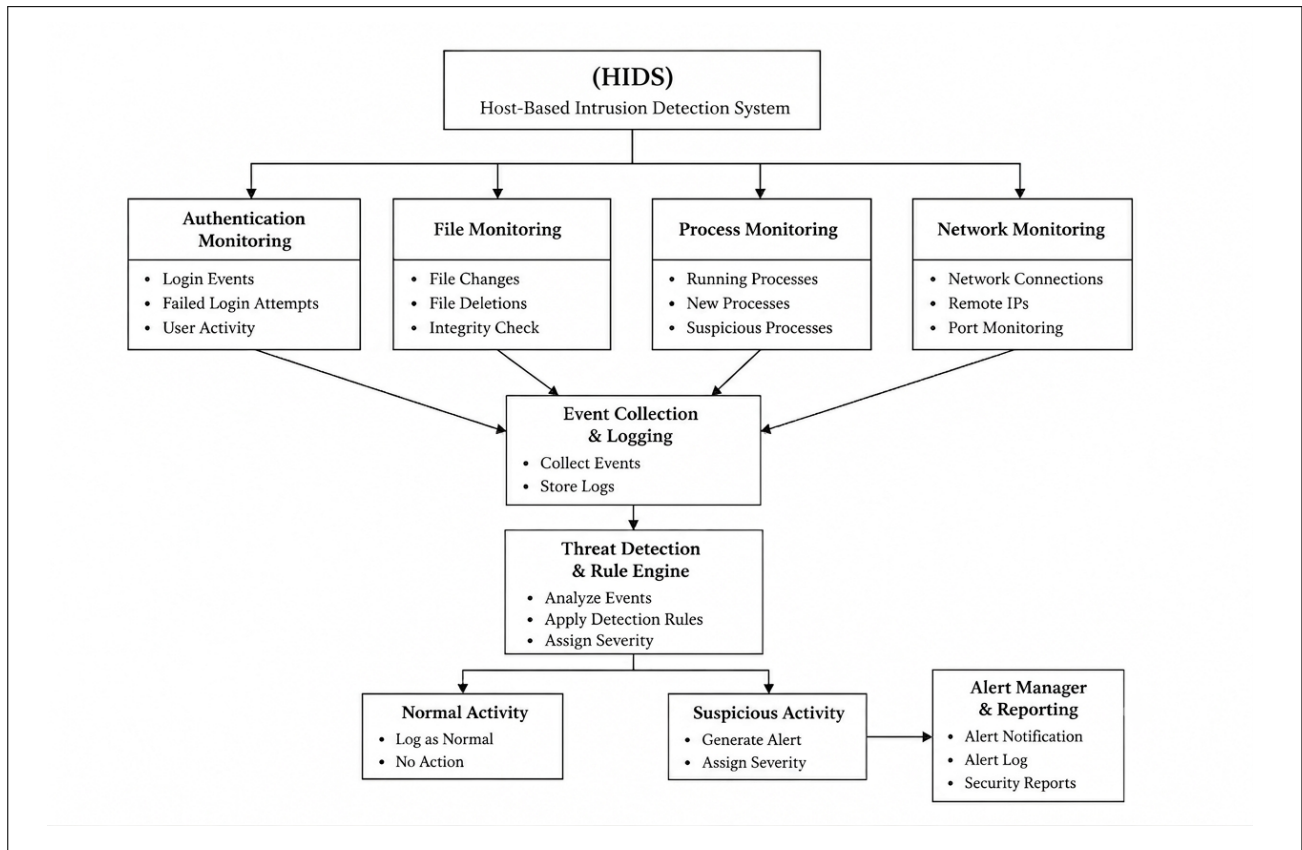
We plan to build the HIDS using C++ and combine Object-Oriented Programming and Data Structures and Algorithms concepts. The system will collect security events related to login activity, files, processes, and network connections. These events will be represented using suitable C++ classes and stored in appropriate data structures.

For OOP, we will create classes such as SecurityEvent, LoginMonitor, FileMonitor, ProcessMonitor, NetworkMonitor, ThreatDetector, AlertManager, and LogManager. Common monitoring functionality can be represented through base classes, while specific monitoring modules can implement their own behavior using inheritance and polymorphism.

For DSA, queues can be used to process incoming security events, priority queues can be used to handle alerts according to severity, and hash maps can be used for quick lookup of users, files, processes, or known entities. Sets can be used to maintain collections of known or suspicious items, while searching and sorting algorithms can help analyze and organize security events.

The main components will then be connected to create a working HIDS where events are collected, analyzed against predefined rules, assigned a severity level, and converted into alerts when suspicious behavior is detected.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

The main outcome of the project will be a working C++ based Host-Based Intrusion Detection System capable of monitoring selected host activities and detecting predefined suspicious behavior. The system will collect security events related to authentication, files, processes, and network activity.

The system will analyze these events using predefined detection rules and generate alerts based on the severity of detected activities. It will also maintain security logs that can be used to review previously detected events.

Through this project, we aim to demonstrate how DSA concepts, algorithms, and OOP principles can be combined to solve a practical cybersecurity problem. The final deliverables will include a functional C++ based HIDS, system architecture and design, source code, test cases/results, and the required project documentation.

Assumptions

We assume that the HIDS will monitor a defined set of host activities such as login events, file changes, processes, and selected network information. Some security events may be simulated instead of being collected from a real operating system. We also assume that predefined rules and thresholds will be used to classify suspicious behavior. The system is intended as an academic prototype and not as a replacement for a complete commercial security solution.

References

- 1 : C++ Documentation and Learning Resources*
- 2 : Object-Oriented Programming course material*
- 3 : Data Structures and Algorithms course material*
- 4 : Operating System and System Security concepts*
- 5 : GeeksforGeeks – Data Structures and Algorithms*
- 6 : Microsoft Documentation – Windows Security and System Monitoring*